Lorenzo Valente

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INSPIRE-HEP | ORCID | GitHub | Hugging Face | LinkedIn | Website

Education

By March 2027	**PhD in Physics** University of Hamburg, Germany **Research Focus**: Machine Learning for Fast Calorimeter Simulation, in collaboration with DESY. Supervisor: Prof. Gregor Kasieczka, co-supervisor Dr. Frank Gaede
March 2023	**Master in Nuclear and Subnuclear Physics** University of Bologna, Italy Thesis title: "A variational autoencoder application for real-time anomaly detection at CMS". Supervisor: Prof. Daniele Bonacorsi
April – August 2021	**Erasmus+ Exchange Semester** Ludwig Maximilian University of Munich, Germany Courses: Machine Learning in Fundamental Physics (S. Krippendorf), Radiation Detectors for Medical Applications.
July 2020	**Bachelor in Physics** University of Bologna, Italy Thesis title: "Reconstruction of non-prompt charmed baryon Λ_c with Boosted Decision Trees technique" (ALICE experiment). Supervisor: Prof. Andrea Alici
October 2018	**Master in Classical Piano** Conservatory of Music Luisa D'Annunzio, Pescara, Italy

Talks

Conference talks

- **European AI for Fundamental Physics Conference (EuCAIFCon)** — **Heidelberg, Germany**
  "Cross-Geometry Shower Simulation via Pre-training and Transfer" — 24 August 2026
- **14th Large Hadron Collider Physics Conference (LHCP)** — **Paris, France**
  "ML and Detector Simulation" (invited) — 21 May 2026
- **23rd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT)** — **Hamburg, Germany**
  "Cross-Geometry Fast Electromagnetic Shower Simulation" — 11 September 2025
- **7th Inter-Experimental LHC Machine Learning Workshop** — **CERN, Switzerland**
  "Cross-Geometry Fast Electromagnetic Shower Simulation" — 20 May 2025

Other talks

- **Glühwein Workshop 2025** — **Aachen, Germany**
  "Cross-Geometry Fast Electromagnetic Shower Simulation" — 15 December 2025
- **8th Round Table on Deep Learning at DESY** — **Hamburg, Germany**
  "Cross-Geometry Transfer Learning in Fast Electromagnetic Shower Simulation" — 14 November 2025
- **DPG Spring Meeting 2025** — **Göttingen, Germany**
  "Adaptive Generative Modeling for Accelerated Calorimeter Simulations via Domain Transfer" — 3 April 2025
- **DPG Spring Meeting 2024** — **Karlsruhe, Germany**
  "Adaptive Generative Modeling for High-Granularity Calorimeters" — 8 March 2024
- **ML-INFN Weekly Meeting** — **INFN, Italy**
  "Joint Variational Auto-Encoder for Anomaly Detection in High Energy Physics" — 17 July 2023

Research Experience

October 2023 – present **Doctoral Researcher**
University of Hamburg, Germany
Machine learning for fast calorimeter simulation, based on generative models and on transfer learning to unseen detector geometries.
Supervisors: Prof. Gregor Kasieczka and Dr. Frank Gaede, in collaboration with Dr. Katja Krüger

April – July 2023 **Summer Student**
Instituto de Astrofísica de Canarias (IAC), Spain
Deep learning classification of heavy nuclei in cosmic rays (iron spectrum), ensemble approach reaching AUC 0.90.
Funded by an Erasmus+ traineeship grant.
Supervisors: Dr. Mónica Vázquez Acosta and Dr. Miguel Molero González

June 2022 – March 2023 **Master's Thesis Research**
University of Bologna, Italy
Variational autoencoder for real-time anomaly detection at the CMS Level 1 trigger on public simulation, quantisation-aware training with QKeras and hls4ml.
Supervisor: Prof. Daniele Bonacorsi

Teaching

AI in Physics, MSc course, University of Hamburg, tutor, 2025

Python tutorial, BSc course, University of Hamburg, tutor, 2025

BSc thesis co-supervision, Investigating the Behavior of Generative Models in Particle Physics under Recursive Training, University of Hamburg, 2025

Schools and Workshops

ELLIS Winter School on Foundation Models, Amsterdam, Netherlands March 2025

Generative Modeling Summer School, Eindhoven, Netherlands July 2024

Machine Learning with FPGAs for Particle Physics, Bologna, Italy November 2022

Quantum Sensing, Information Processing and Computing, Bologna, Italy July 2022

Technical Skills

Simulation & HEP Software: ROOT, Geant4, DD4hep (ddsim, k4geo, ddfastsim), Key4hep, LCIO, uproot, awkward

Programming Languages: Python, C/C++, Bash, LabVIEW

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, ONNX export, mixed-precision training, experiment tracking (Comet)

Computing Infrastructure: Slurm (DESY Maxwell), HTCondor (DESY NAF), Singularity containers, distributed multi-GPU training, HDF5

Real-Time & FPGA: hls4ml, QKeras, Vivado HLS, VHDL

Software Engineering: Git, CI, pytest, Docker

Languages

Italian (Native), English (Fluent), Spanish (B2), German (B1)

Lorenzo Valente

Publications

Publications

* *corresponding author*

Journal articles and preprints

- F. Gaede, G. Kasieczka, **L. Valente***, *Cross-Geometry Transfer Learning in Fast Electromagnetic Shower Simulation*, JINST **21** (2026) P07037.
  Contribution: conceived and implemented the transfer framework, ran all trainings and analysis, wrote the paper.
- H. Day-Hall*, ..., **L. Valente**, *CaloClouds3: Ultra-fast Geometry-Independent Highly-Granular Calorimeter Simulation*, JINST **21** (2026) P03018.
  Contribution: trained and tuned successive model versions, fed improvements from my cross-geometry transfer work back into the model, contributed figures and manuscript review.
- T. Buss, ..., **L. Valente***, *Transferable Fast Calorimeter Shower Generation via Multi-Geometry Pre-training*, JINST **21** (2026) P10041, alphabetical authors.
  Contribution: led the study design, dataset production, training campaigns and analysis, wrote the paper.

Conference proceedings

- T. Buss, ..., **L. Valente***, *Point-Cloud Generative Models for Fast Calorimeter Simulation across Particles and Geometries*, arXiv:2609.30403 (2026), LHCP 2026, accepted in PoS.
  Contribution: gave the invited talk and wrote the paper.
- T. Buss, ..., **L. Valente***, D. Weber, *Cross-Geometry Transfer and Model Collapse in Point Cloud Calorimeter Shower Generation*, arXiv:2609.28661 (2026), ACAT 2025, submitted to PoS.
  Contribution: gave the talk and wrote the paper, the model collapse study comes from the BSc thesis I co-supervised.
- T. Buss, ..., **L. Valente**, *Jevons' Paradox and Fast Generative Simulation for HEP: Why Realistic Benchmarking is Essential*, arXiv:2609.24306 (2026), SciPost Phys. Proc. (submitted).
  Contribution: co-author through CaloClouds3, the model the article builds on, which I trained and tuned across its versions.
- **L. Valente*** et al., *Joint Variational Auto-Encoder for Anomaly Detection in High Energy Physics*, PoS **ISGC&HEPiX2023** (2023) 014.
  Contribution: developed the model, ran the trainings and the analysis and wrote the paper, MSc work on real-time anomaly detection at CMS.